FC01 compact two-nozzle filling cell

Siemens-authoritative controls and bounded NVIDIA quality architecture

FICTIONAL ENGINEERING PROJECT — NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE — REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Release status	What is implemented	What remains blocked
PARTIALLY COMPLETE	Canonical engineering model; selected hardware; modular Siemens-oriented SCL; HMI and drive specifications; OPC UA PLC-AI contract; deterministic two-channel simulator; 32 scenarios; schedules and manifest.	Native TIA/WinCC compile and archive; Startdrive; PLCSIM; revision-B QET and CAD; real dataset; TAO/DeepStream/TensorRT execution; model metrics and latency.

This report is an evidence index. It is not proof of construction readiness, functional safety, native compile, FAT, SAT, electrical test, physical commissioning or AI performance.

Integrated architecture and deterministic ownership

The PLC owns all machine sequence, interlocks, timeouts and transfer permission. NVIDIA is a standard quality-inspection device and cannot perform or bypass safety functions.

Layer	Owner	Deterministic responsibility	Failure behavior
Field / power	Electrical	Identified device, cable/core, terminal and I/O channel	Fail-closed valves; external safety removes hazardous energy
Equipment control	S7-1500 FB instances	Gate, clamp, two fill channels, conveyor/pump VFD, capper	Module timeout/contradiction fault; outputs safe
Machine coordinator	S7-1500	Legal states, accepted command edges, no automatic restart	HOLDING, CONTROLLED_STOPPING or FAULTED
Operator interface	MTP700 Unified	Role-checked handshake; feedback and disabled reason	No direct physical-output writes
Quality vision	Jetson / DeepStream	Acquire, infer and publish correlated result	Timeout/stale ID/low confidence/fault -> quality hold
Regression	Python simulator	Interface and sequence fault injection	Design evidence only; never PLCSIM proof

Machine states: UNINITIALIZED, INITIALIZING, STOPPED, READY, AUTOMATIC, MANUAL_SETUP, HOLDING, CONTROLLED_STOPPING, FAULTED and RECOVERY_RESET. Reset returns to STOPPED and never initiates motion.

Selected Siemens and edge hardware baseline

Function	Selected item	Order number / status	Verification boundary
PLC	SIMATIC S7-1500 CPU 1511-1 PN	6ES7511-1AL03-0AB0	Current official lifecycle checked; V20 device catalog/compile open
Pulse counting	TM Count 2x24V	6ES7550-1AA01-0AB0	Two independent 24 V pulse channels; TIA configuration open
DI / DO / AI	DI 32 HF; DQ 32 HF; AI 8xU/I HF	6ES7521-1BL00-0AB0; 6ES7522-1BL01-0AB0; 6ES7531-7NF00-0AB0	Front connectors, wiring and catalog compatibility open
HMI	MTP700 Unified Comfort	6AV2128-3GB06-0AX1	WinCC V20 compile/licence open
Drives	Two SINAMICS G120C PN 0.75 kW	6SL3210-1KE12-3UF2	Ratings provisional; Startdrive absent
Vision	Jetson Orin NX 16GB on industrial carrier	Carrier/integration SKU TBD	Carrier, thermals, EMC, camera and real dataset open

Authoritative engineering target: TIA Portal V20, executable/product version 2000.0.9501.1. Installed STEP 7 and WinCC V20 components do not prove usable licence entitlement. Current identity is not authorized for TIA Openness. Startdrive and PLCSIM are absent.

PLC-AI acceptance contract

Transport selection: OPC UA across a routed quality zone. The PLC publishes an atomic request and one-shot trigger. Results are accepted only with the current ID, fresh heartbeat, ready state, valid flag and non-contradictory payload.

PLC -> NVIDIA	NVIDIA -> PLC	Acceptance rule
VISION_ENABLE; INSPECTION_TRIGGER; INSPECTION_ID; RECIPE_ID; EXPECTED_BOTTLES; TARGET_FILL_LEVEL; PLC_HEARTBEAT	VISION_READY; VISION_BUSY; RESULT_VALID; RESULT_ID; BOTTLE_1_PASS; BOTTLE_2_PASS; FILL_1_STATUS; FILL_2_STATUS; LEAK_OR_SPILL_DETECTED; LOW_CONFIDENCE; VISION_WARNING; VISION_FAULT; INFERENCE_TIME; VISION_HEARTBEAT	RESULT_ID equals current INSPECTION_ID; heartbeat fresh; both bottle passes; fill statuses in-range; no leak, low confidence or fault. Otherwise HOLDING.

No model is delivered. No training, ONNX export, TensorRT engine, DeepStream execution, metric, confusion matrix, false-accept/false-reject analysis or latency benchmark is claimed. The available RTX 5060 Laptop GPU does not substitute for the absent CUDA/TAO/DeepStream stack or representative dataset.

Validation evidence and release blockers

Gate	Evidence	Status
Canonical consistency	Automated checks cover required structure, unique addresses/symbols, exact AI interface, CSV parse, derivation and native baseline hashes	PASS
Deterministic regression	32 required scenarios; all blocking cases keep pump and both valves off; only normal cycle releases; no automatic restart	PASS - not PLCSIM
Workbook	16 sheets rendered; formula-error scan returned zero matches; contact-sheet and detailed visual review	PASS
QElectroTech	Audited inherited hash; well-formed XML; retained native reopen evidence for Rev A	REVISION-B GATE OPEN
Panel CAD	Audited inherited FCStd/STEP/IGES/DXF hashes and retained Rev-A reopen/reimport evidence	REVISION-B GATE OPEN
TIA / WinCC / drives	V20 installed; no native project, compile, cross-reference, archive restore or Startdrive evidence	BLOCKED
PLCSIM	Not installed	BLOCKED
NVIDIA model/runtime	Hardware detected; runtime stack and dataset absent	BLOCKED

Release use: controlled design review and continuation in qualified native tools only. Read release/RELEASE_NOTES.md, release/manifest.json, 14_qa/acceptance_gate_status.md and 00_project_control/open_issues.csv before work.